

electron/git-service.cjs

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Overview

`electron/git-service.cjs` is a new CommonJS module that bundles synchronous Git helpers for the Electron renderer. It centralises Git interactions that were previously scattered across the codebase.

Key changes

- **Imports** (R3-R6): `node:fs` , `node:path` , `node:os` , `node:child_process` .
- **Command wrappers** (R12-R36): `runGit` and `tryRunGit` use `spawnSync` to execute Git, surface errors, and trim stdout.
- **Repository introspection** (R38-R120): `isGitRepo` , `getRepoStatus` , and `parsePorcelainStatus` collect branch, upstream, file status, and remote data.
- **Common Git ops** (R137-R154): `stageAll` , `commit` , and `push` expose `git add -A` , `git commit` , and `git push` .
- **Worktree utilities** (R156-R182): `ensureWorktreeBase` , `removeDirSafe` , `checkoutRefWorktree` , and `cleanupNovadiffWorktrees` manage temporary detached worktrees for PR comparisons.
- **PR comparison setup** (R191-R205): `preparePrCompareRoots` creates left/right roots for a PR diff.
- **Tooling detection** (R207-R214): `detectGitTooling` reports Git availability and version.
- **Exports** (R216-R229): All helpers are exported for external use.

Impact

- **Error handling**: `runGit` throws on non-zero exit or spawn errors; callers must catch these.
- **Synchronous execution**: `spawnSync` blocks the event loop; acceptable for small scripts but may stall the UI during heavy Git operations.
- **Node compatibility**: Relies only on Node 14+ core modules; no external dependencies.
- **Observability**: Errors surface with messages like `git exited X` or `Failed to run git` .

Risks & follow-ups

1. **Git absence** – `detectGitTooling` must be checked before using any helper; otherwise, an exception will crash the renderer.
2. **Worktree cleanup** – `cleanupNovadiffWorktrees` should run on app exit to avoid accumulating temp directories.
3. **Path sanitization** – `checkoutRefWorktree` sanitises labels but still uses `path.basename` ; verify no edge-case injection.

4. **Build inclusion** – Ensure the new `.cjs` file is referenced in the Electron build config; otherwise, runtime imports will fail.